同行评审意见逐条翻译对照表（全量完整版）

Full Manuscript Peer Review Comments: English-to-Chinese Unabridged Parallel Translation

Reviewer 3

日期：16 Oct 2026 | 07:53

Comment #1

The phrase "Impact of" in the title may imply a cause-and-effect relationship. However, this study employs a non-randomized retrospective cohort design, with patients grouped based on the characteristics of the interventions they received.

标题中的“Impact of”（……的影响）一词可能会暗示因果关系。然而，本研究采用了非随机回顾性队列设计，患者是根据其接受的干预特征进行分组的。

Comment #2

In the results section of the abstract, numerical data are provided; however, there are inconsistencies in the interpretation of readmissions and the benefits of the intervention, in the conclusion, claims regarding improvements in renal and metabolic outcomes should be qualified, taking into account the study design and the results of the analysis.

在摘要的结果部分提供了数值数据；然而，在对再入院率和干预获益的解释上存在不一致之处；在结论部分，考虑到研究设计和分析结果，关于肾脏和代谢结局改善的表述应当更加审慎和有限制性。

Comment #3

The author needs to provide supporting literature for the identified gaps (clinical gap: it remains unclear whether varying intervention intensities are associated with differences in outcomes for patients with CKD; and measurement or methodological gap: it is unclear how intervention intensity can be consistently quantified using various clinical pharmacy service indicators).

作者需要为所提出的研究空白提供支持文献（临床空白：尚不清楚不同干预强度是否与 CKD 患者结局的差异相关；测量或方法学空白：尚不清楚如何利用各种临床药学服务指标来一致性地量化干预强度）。

Comment #4

The author uses K-means to form intensity clusters; it should be explained why clustering is more appropriate than using composite scores or pre-defined categories.

作者使用 K-means 聚类来形成强度聚类；应当解释为什么聚类比使用综合评分或预先定义的类别更为合适。

Comment #5

It is advisable for the author to conclude the introduction with a single paragraph that connects four elements: what remains unknown from previous research, how this study differs in its approach, the expected contribution relative to existing evidence, and the research question or hypothesis to be tested.

建议作者在引言末尾用一个段落串联四个要素：以往研究中尚未解决的问题、本研究在方法上的创新之处、相对于现有证据的预期贡献，以及待检验的研究问题或假说。

Comment #6

The methodology is well described; however, several points require attention: there is insufficient detail regarding the reasons for the exclusion of 51 patients; since patients participated in the multidisciplinary program on a voluntary basis, the potential for selection bias should be addressed; the age differences between groups suggest that patient characteristics might be related to the intensity of the intervention; and while the diagnostic criteria for DKD refer to external sources, their operational implementation needs to be explained adequately.

方法学部分描述详尽；但仍有几个点需要注意：排除 51 名患者的具体原因细节不足；由于患者是自愿参与多学科项目的，应当探讨潜在的选择性偏倚；组间的年龄差异提示患者特征可能与干预强度相关；虽然 DKD 的诊断标准参考了外部文献，但其具体的操作实施方式需要做出充分解释。

Comment #7

In general, the selection of outcomes is appropriate. However, several aspects require clarification: Is the primary outcome being examined the value at the sixth month, the change from baseline, or the pattern of change over the follow-up period? How did the authors handle data that were unavailable at the third or sixth month? Were laboratory measurements performed using consistent procedures throughout the study period? Why was an eGFR improvement threshold of 12.5% chosen for the regression analysis? Is the definition of MACE used consistent throughout the manuscript?

总体而言，结局指标的选择是恰当的。然而，有几个方面需要澄清：所考察的主要结局是第 6 个月时的数值、较基线的变化值，还是随访期间的变化趋势模式？作者是如何处理第 3 个月或第 6 个月缺失的数据的？在整个研究期间，实验室测量是否采用了统一的操作规程？为什么在回归分析中选择 12.5% 的 eGFR 改善幅度作为阈值？整篇文稿中对 MACE（主要不良心血管事件）的定义是否保持一致？

Comment #8

The results indicate that the high-intensity group had a higher eGFR and lower UACR and HbA1c levels at six months. While these differences can be interpreted in the discussion, one should not immediately conclude that the intensity of the intervention caused the improvement. It is necessary to distinguish between differences in final values and differences in changes from baseline.

结果显示，高强度组在 6 个月时的 eGFR 较高，UACR 和 HbA1c 水平较低。虽然这些差异可以在讨论部分进行解释，但不应立即得出“干预强度导致了改善”的结论。有必要区分“终点值的差异”与“较基线变化值的差异”。

Comment #9

Regarding the conclusion, the following points should be noted:
(1) The conclusion does not sufficiently distinguish between significant and non-significant results. Differences in renal and metabolic indicators at six months were reported as significant, whereas differences in readmission, dialysis, and MACE (major adverse cardiovascular events) were not statistically significant.
(2) Claims regarding long-term benefits are not sufficiently supported, as the observation period lasted only six months. Therefore, statements concerning improvements in long-term outcomes should be framed as potential implications rather than as proven benefits.
(3) The methodological novelty should be presented with appropriate perspective. K-means is an established method; the potential novelty lies in its application specifically, using it to cluster patterns of clinical pharmacy interventions for patients with DKD (diabetic kidney disease) rather than in the algorithm itself.

关于结论部分，应注意以下几点：
（1）结论未充分区分具有统计学显著性与不显著的结果。6 个月时的肾脏和代谢指标差异被报告为显著，而再入院率、透析和 MACE（主要不良心血管事件）的差异在统计学上并不显著。
（2）关于长期获益的断言缺乏充分支持，因为观察期仅持续了 6 个月。因此，关于长期结局改善的表述应归框为“潜在影响”，而非“已证实的获益”。
（3）方法学的新颖性应予以客观中肯的表述。K-means 是一种成熟的技术方法；其潜在的新颖性在于将其专门应用于糖尿病肾病（DKD）患者临床药学干预模式的聚类分析，而非算法本身。

Reviewer 2

日期：21 Aug 2026 | 12:46

Overview / General Assessment

As the role of clinical pharmacists in multidisciplinary care continues to expand, determining how pharmacist services should be differentiated according to patients' clinical needs has become an increasingly important question. This manuscript addresses this issue from a perspective that extends beyond simply evaluating whether pharmacist intervention is effective. By attempting to conceptualize and quantify intervention intensity, the study explores whether different levels and patterns of pharmaceutical care are associated with differential renal, metabolic, and clinical outcomes. This is a potentially valuable approach, particularly for informing the more efficient allocation and tailoring of pharmacist services. However, several issues require clarification.

随着临床药师在多学科协作诊疗中的作用不断扩大，如何根据患者的临床需求对药师服务进行差异化配置已成为一个日益重要的课题。本稿件从一个超越“仅评估药师干预是否有效”的视角探讨了这一问题。通过尝试将干预强度概念化和定量化，本研究探讨了不同水平和模式的药学服务是否与肾脏、代谢及临床结局的差异相关。这是一种具有潜在价值的方法，特别是在为更高效地配置和量身定制药师服务提供依据方面。然而，有几个问题需要进一步澄清。

Question 1

1. The central contribution of this study lies in its attempt to quantify pharmacist-driven care, incorporating both clinical activities and management-oriented activities, and to relate these intervention patterns to patient outcomes. However, the construction and operationalization of intervention intensity require substantially greater clarification.

(1) Why were these specific indicators selected to characterize intervention intensity? The manuscript does not clearly indicate whether their selection was guided by an explicit theoretical framework, previous empirical evidence, clinical consensus, or data availability. Given that the validity of the subsequent classification depends fundamentally on the appropriateness of these indicators, a clear rationale for their inclusion is needed.

(2) The operational definition of each indicator should be described in greater detail. For example, was therapeutic regimen optimization coded as a binary event, quantified by the number of interventions, or assessed using another approach? Similarly, was physician acceptance of pharmacist recommendations expressed as the number of accepted recommendations, the proportion of accepted recommendations, or another metric? Without explicit definitions, it is difficult to assess the comparability and interpretability of the individual components contributing to intervention intensity.

(3) PCA was used for dimensionality reduction before clustering, but the relationship between the original variables, the principal components, and the variables subsequently reported in the regression analyses is not sufficiently transparent. In particular, the definitions of categories such as "medium-to-high adherence" and "high medication education quantity" (Factors Associated with eGFR Improvement and Correlation Analysis section) should be clarified. The authors should specify whether these categories were predefined, derived from empirical cutoffs, informed by clustering results, or based on clinically meaningful thresholds. A clear description of the complete data-processing pipeline would substantially improve methodological transparency and interpretational consistency.

1. 本研究的核心贡献在于尝试对药师主导的照护（包含临床活动和管理导向活动）进行定量化，并将这些干预模式与患者结局联系起来。然而，干预强度的构建和具体操作化过程需要大幅增加说明。

（1）为什么选择这些特定指标来表征干预强度？文稿未明确说明其选择是基于明确的理论框架、以往的实证证据、临床共识还是数据的可获得性。鉴于后续分类的有效性根本上取决于这些指标的恰当性，因此需要提供明确的纳入理由。

（2）每个指标的操作性定义应更详细地描述。例如，治疗方案优化是被编码为二元事件、按干预次数定量，还是采用其他方法评估？同样，医师对药师建议的采纳率是用采纳建议的数量、采纳建议的比例还是其他指标表示？若没有明确的定义，很难评估构成干预强度的各个组成部分的可比性和可解释性。

（3）在聚类前使用了主成分分析（PCA）进行降维，但原始变量、主成分以及随后在回归分析中报告的变量之间的关系不够透明。特别是对“中高依从性”和“高用药教育数量”等类别的定义（见“eGFR改善的相关因素及相关性分析”部分）应予以澄清。作者应说明这些分类是预先定义的、源自实证切点、受聚类结果启发，还是基于具有临床意义的界值。对完整数据处理流程的清晰描述将大大提高方法学透明度和解释一致性。

Question 2

2. Since all participants were enrolled in an MDT program, the level of pharmacist involvement may have been influenced by patients' baseline clinical complexity. Patients with more severe disease, poorer medication adherence, or more complicated treatment regimens may have been more likely to receive intensive pharmacist intervention. This raises an important concern regarding confounding by indication. Moreover, if intervention intensity was determined during the same period in which clinical outcomes evolved, the temporal ordering between exposure and outcome should also be clarified to exclude potential reverse causality. The authors should therefore describe how baseline disease severity and other determinants of intervention allocation were accounted for in the analysis.

2. 由于所有参与者都加入了多学科团队（MDT）项目，药师的参与程度可能受到了患者基线临床复杂性的影响。病情较重、用药依从性较差或治疗方案较复杂的患者可能更有可能接受密集的药师干预。这引发了关于“适应症混杂”（confounding by indication）的重要关注。此外，如果干预强度是在临床结局演变的同一时期内确定的，则还应澄清暴露与结局之间的时序关系，以排除潜在的反向因果关系。因此，作者应说明在分析中是如何控制基线疾病严重程度及其他影响干预分配的因素的。

Question 3

3. The covariate strategy used in the regression analyses requires further clarification. The authors report adjustment for "gender, age, history of diabetes, clinical staging, disease severity, and intervention approaches," but several issues remain unclear: (1) the coding and functional form of these variables should be explicitly reported. For example, was age modeled continuously or categorically? Were clinical staging and disease severity treated as ordinal, categorical, or binary variables? (2) the term "history of diabetes" is potentially ambiguous in a cohort consisting entirely of patients with diabetic kidney disease. The authors should clarify whether this variable refers to diabetes duration or another diabetes-related characteristic. (3) several clinically relevant confounders appear to have been omitted from the model, including comorbidity burden, number of medications, and insulin use, all of which are likely to influence both pharmacist intervention intensity and patient outcomes. The authors should clarify these definitions and justify their covariate selection strategy.

3. 回归分析中使用的协变量策略需要进一步澄清。作者报告调整了“性别、年龄、糖尿病史、临床分期、疾病严重程度和干预方式”，但仍有几个问题不明确：（1）应明确报告这些变量的编码和函数形式。例如，年龄是作为连续变量还是分类变量建模的？临床分期和疾病严重程度是被视为等级变量、无序分类变量还是二元变量？（2）在一个完全由糖尿病肾病患者组成的队列中，“糖尿病史”一词可能存在歧义。作者应澄清该变量是指糖尿病病程还是其他与糖尿病相关的特征。（3）模型中似乎遗漏了几项具有临床相关性的混杂因素，包括合并症负担、用药数量和胰岛素使用情况，这些因素都很可能同时影响药师干预强度和患者结局。作者应澄清这些定义并解释其协变量选择策略的合理性。

Question 4

4. The study initially assessed 148 patients but ultimately included only 97, corresponding to the exclusion of approximately 34.5% of the initial sample. However, the reasons for these exclusions are not sufficiently described. The manuscript should provide a clear accounting of the exclusion process, including the number of patients excluded for each specific reason (e.g., ineligibility, missing data, absence of pharmacist intervention, insufficient follow-up, or other causes). A participant flow diagram would further improve the transparency of the study population.

4. 本研究最初评估了 148 名患者，但最终仅纳入了 97 名，相当于排除了初始样本的约 34.5%。然而，排除这些患者的原因描述得并不充分。文稿应清晰交代排除过程，包括因每个具体原因被排除的患者数量（例如：不符合入组条件、数据缺失、缺乏药师干预、随访不足或其他原因）。参研流程图（Flow diagram）将进一步提高研究人群的透明度。

Question 5

5. Several statements in the Results section extend beyond direct reporting of the observed findings and instead imply mechanisms or explanatory pathways. For example, the statement that "medication education functioned as a multifunctional mediator... (page7 lines 164-166)" appears to invoke a mediation mechanism that was not formally tested. Similarly, the assertion that "more rational baseline regimens necessitate fewer adjustments... (page7 lines 167-168)" represents an interpretation rather than a direct empirical result. Statements that infer mechanisms, mediation, or causal explanations beyond the analyses presented should be moved to the Discussion section and supported by appropriate references or additional analyses. The Results section should remain focused on the findings generated by the prespecified analyses.

5. “结果”部分中的若干表述超出了对观察到的研究发现的直接报告，而是暗示了机制或解释性路径。例如，“用药教育起到了多功能中介作用……（第7页第164-166行）”这一表述似乎引入了未经正式检验的中介机制。同样，“更合理的基线方案需要的调整更少……（第7页第167-168行）”这一断言代表的是一种解释，而非直接的实证结果。超出所呈现分析范围、推断机制、中介作用或因果解释的表述应移至“讨论”部分，并辅以适当的参考文献或补充分析。“结果”部分应保持专注于预先指定的分析所产生的结果。

Question 6

6. Two aspects of the outcome analysis warrant clarification:
(1) the manuscript identifies eGFR, UACR, and HbA1c as "primary outcomes." The use of multiple primary outcomes requires a clearer conceptual and statistical justification. The authors should clarify whether these were prespecified as co-primary outcomes and explain how the issue of multiple comparisons was addressed. Alternatively, if the study was exploratory, the distinction between primary and secondary outcomes should be framed accordingly. In either case, the hierarchy and clinical rationale for the selected outcomes should be made explicit.
(2) the analysis appears to compare raw 6-month outcome values between groups. Even when baseline differences are not statistically significant, comparisons based solely on follow-up values may not adequately account for individual baseline variability. At minimum, follow-up outcomes should be analyzed with adjustment for their corresponding baseline values. Alternatively, a repeated-measures or mixed-effects approach could make fuller use of the longitudinal data by directly examining changes over time and potential group-by-time interactions. Such approaches would provide a more appropriate estimate of the association between intervention intensity and outcome trajectories.

6. 结局指标分析有两个方面需要澄清：
（1）文稿将 eGFR、UACR 和 HbA1c 均列为“主要结局”。使用多个主要结局需要更明确的概念和统计学依据。作者应澄清这些指标是否预先指定为共同主要结局（co-primary outcomes），并解释是如何处理多重比较问题的。或者，如果研究属于探索性研究，则应相应地界定主要结局与次要结局的区别。无论哪种情况，所选结局的层级结构和临床合理性都应予以明确说明。
（2）分析似乎是在比较各组之间 6 个月时的原始结局数值。即使基线差异在统计学上无显著性，仅基于随访值的比较也可能无法充分控制个体基线变异。至少应在调整其相应基线值的情况下对随访结局进行分析。或者，重复测量或混合效应模型可以通过直接考察随时间变化的情况以及潜在的“组别-时间”交互作用，更充分地利用纵向数据。这些方法将对干预强度与结局轨迹之间的关联提供更恰当的估计。

Question 7

7. Several additional limitations should be acknowledged more explicitly.

First, given the retrospective observational design, the findings should be interpreted as associations rather than evidence of causal effects.

Second, reducing a multidimensional pattern of pharmacist activities to a binary high-versus-low intensity classification may obscure heterogeneity in the underlying intervention profiles, particularly when the specific components driving group membership remain insufficiently defined.

Third, the intervention indicators and classification thresholds were derived from a single-center sample and have not been externally validated.

7. 应当更明确地承认其他几项局限性。

首先，鉴于回顾性观察性研究设计，研究发现应解释为关联性，而非因果效应的证据。

其次，将药师多维度的活动模式简化为“高强度 vs 低强度”的二元分类，可能会掩盖潜在干预特征的异质性，尤其是当驱动分组的具体构成要素定义仍不够明确时。

第三，干预指标和分类切点来源于单中心样本，尚未经过外部验证。